Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			A **and** V correctly added to circuit Don't accept a	1			1		1
	(b)	(i)		All 5 points plotted correctly ignore missing plot at (0,0) within <1 small square (2) 4 points correctly plotted within <1 small square (1) 3 or less points correctly plotted within <1 small square (0) Straight line of best fit through (0,0) (1)		3		3	3	3
		(ii)		0.08 [A] **ecf or** value from candidate's graph tolerance of ± 0.004		1		1	1	1
		(iii)		$\frac{4}{0.08\ \textbf{ecf}}$ (1) Resistance = 50 [Ω] (1)	1	1		2	2	2
		(iv)		Line is straight [through the origin] <u>so agree</u> **OR** as V doubles, I doubles <u>so agree</u> **OR** constant gradient <u>so agree</u> **OR** as V increases, I increases at a steady rate <u>so agree</u> Accept another calculated value shown to be the same. Accept the line is not straight **ecf** so <u>I disagree</u> Don't accept positive correlation **OR** as V increases, I increases			1	1		1
		(v)		R_{total} = 25 + 50 **(ecf)** (1) R_{total} = 75 [Ω] (1)	1	1		2	2	
				Question 3 total	**3**	**6**	**1**	**10**	**8**	**8**